

Null Hypothesis Significance Testing (NHST)

From Exercise

1. $H_0: \mu = 7$ $H_1: \mu \neq 7$

$t_{cal} = -0,854$ $n = 15$

$p\text{-value} = P(T > |t|) + P(T < -|t|)$ or $2 \cdot P(T > |t|)$
 $= 2 \cdot P(T > 0,854)$

$= (2)(0,20374) = 0,40748 > \alpha = 0,05$

Do not reject H_0

2. $H_0: \mu \geq 5$ $H_1: \mu < 5$

$t_{cal} = -1,687$ $n = 15$

$p\text{-value} = Pr(T \leq -1,687) = 0,05187 > \alpha = 0,05$

Do not reject H_0

3. $H_0: \mu \leq 1,500$ $H_1: \mu > 1,500$

$t_{cal} = 13,664$ $n = 30$

$p\text{-value} = Pr(T > 13,664) = 1,816 \times 10^{-14} < \alpha = 0,05$

Reject H_0

